AI-Driven Academic Risk & Dropout Early-Warning System

Domain: Software Digital Twins / AI & Machine Learning

Frameworks : FastAPI, React, PyTorch/Scikit-Learn

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Overview

1 Problem Statement

2 Abstract & Proposed Solution

3 System Architecture & Flow

4 Implementation

5 Application Domain

6 Expected Outcome

Problem Statement

Existing Problems

- Traditional academic systems identify weak students only after semester results.
- Student data is stored as static records without tracking progress over time.
- Rule-based systems (e.g., attendance below 75%) are inaccurate and cannot explain why a student is at risk.
- Institutions lack an intelligent system for early intervention.

Objective

To design an event-driven, software-only Digital Twin architecture that continuously ingests raw academic and behavioral events, maintains time-series state memory for every student, and executes a trained ML pipeline to predict dropout risk with high recall and mathematical explainability.

Abstract & Proposed Solution

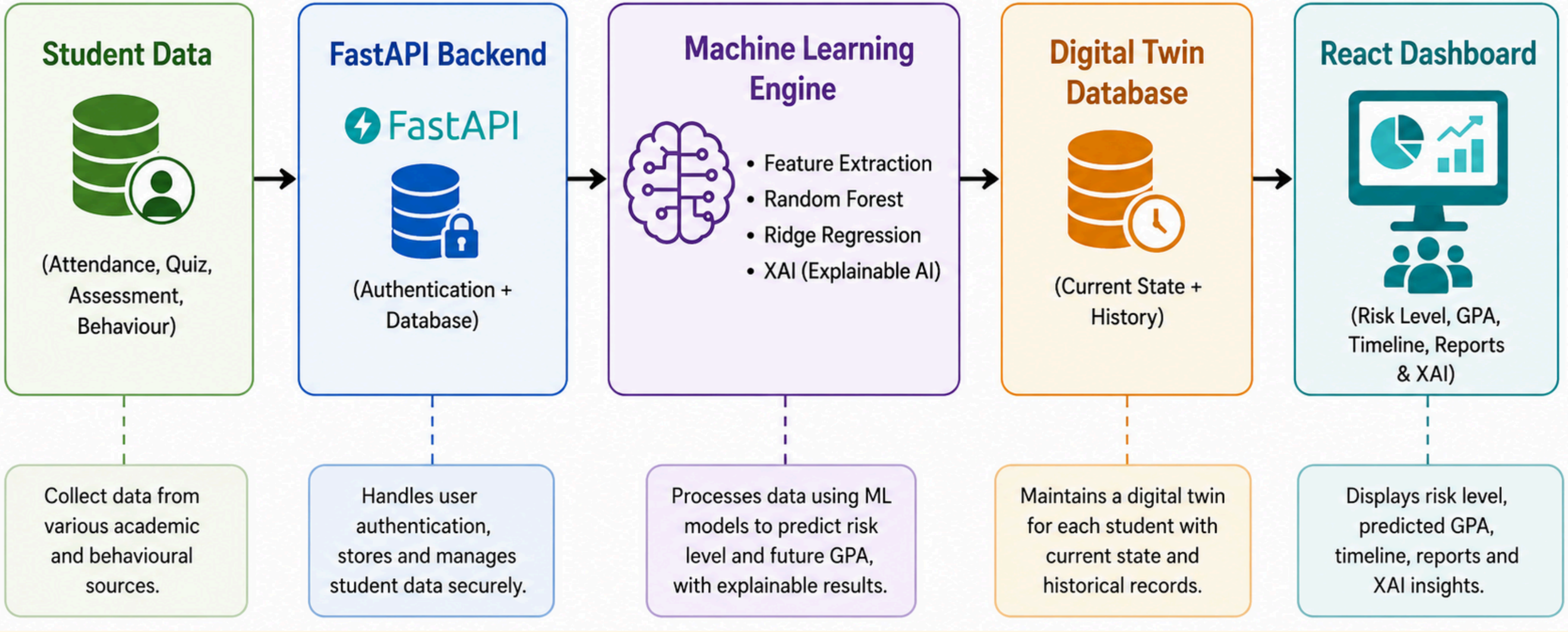
Abstract

- Digital Class Twin is an AI-powered academic monitoring system based on the Digital Twin concept. The system collects student data such as attendance, quiz scores, assessments, and behavioral records. Machine Learning models analyze this data to predict student risk levels and future GPA. The system enables early identification of at-risk students and supports timely academic intervention through an interactive dashboard.

Proposed Solution

- Collect student academic data from multiple sources.
- Store data securely using a FastAPI backend.
- Apply Machine Learning models for risk level and GPA prediction.
- Maintain a Digital Twin for continuous student monitoring.
- Display results through a React dashboard with explainable AI.
- Support teachers and counselors in making early intervention decisions.

System Architecture & Flow



Implementation

Event Ingestion Layer

Attendance App • Assessment Portal • Behavioral Logging System

Immutable raw events

FastAPI Backend & Event Store

JWT Bearer Auth & RBAC • SQLAlchemy Event Store (immutable events table)

Event stream trigger

ML Engine & Analytics Pipeline

Feature Extractor → Risk Classifier → GPA Regressor → Confidence Calculator → XAI Engine

State snapshot write

Digital Twin Persistence Layer

Current Twin State • Time-Series History • Closed-Loop Intervention Log

REST API / data read

Presentation Dashboard (React)

Risk Heatmap • Twin Timeline • XAI Breakdown • PDF Exporter

Technology Stack

Layer	Technologies Used	Key Packages / Specification
Front-End	React 18, Vite, Vanilla CSS, Recharts / Chart.js, Lucide Icons	Single-page application, component architecture
Backend Framework	Python 3.10+, FastAPI, Uvicorn	Async ASGI server, OpenAPI / Swagger docs
Database & ORM	SQLite / PostgreSQL, SQLAlchemy	Relational DB, time-series schema logging
ML Engine	scikit-learn, NumPy, Pandas, Joblib	Random Forest, Ridge Regression, softmax inference
Security & Privacy	PyJWT, Passlib (Bcrypt), OAuth2 Bearer Tokens	Role-based access control (Teacher / Counselor / Admin)

Application Domain

Digital Class Twin can be implemented in:

- Schools - Monitor attendance, performance, and learning progress.
- Colleges & Universities - Identify academically at-risk students and improve retention.
- Online Learning Platforms - Analyze learner engagement and predict performance.
- Academic Counseling Centers - Support counselors with AI-driven student insights.
- Educational Management Systems (ERP/LMS) - Integrate with existing academic management platforms.
- Educational Administrators - Enable data-driven decision making through dashboards and analytics.
- Educational Research & Analytics - Study student learning patterns and academic behavior.
- AI-Based Smart Campus Solutions - Enhance digital transformation in educational institutions.

Expected Outcome

- Early identification of academically at-risk students before examinations.
- Accurate prediction of student risk levels (Low, Medium, and High) and future GPA using Machine Learning.
- Continuous monitoring of student academic performance through Digital Twin technology.
- Explainable AI (XAI) to provide clear insights into the factors influencing each prediction.
- Timely academic intervention by teachers and counselors to improve student performance and reduce dropout risk.
- Data-driven decision making for educational institutions through real-time academic analytics.
- A scalable, intelligent, and reliable academic monitoring system suitable for schools, colleges, and universities.